TXC10-GP-DEGO

Credit Application Governance Analysis

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Data Quality Assessment

Data Quality Issues Six Dimensions

1. UNIQUENESS

Duplicate identifiers (1.2%)

- 2 duplicate **_id** fields (resubmissions)
- 4 records with overlapping SSNs across different applicant names

2. COMPLETENESS

Missing/invalid values (2.8%)

- 6 records missing KYC (**SSN**, **DoB**)
- 2 missing emails & 2 missing zip codes
- 5 **annual_salary** schema drift keys

3. CONSISTENCY

Format & type mixing (23.1%)

- Gender encoded inconsistently: Male / Female vs M / F (21.7%)
- **annual_income** stored as string instead of numeric (1.4%)

4. VALIDITY

Structural violations (31.9%)

- 156 records (31.1%) used non-standard slash-delimited dates
- 4 emails present but failed RFC 5321 structure checks

5. ACCURACY

Impossible values (0.6%)

- 2 Negative **credit_history_months** detected
- 1 Negative **savings_balance** detected

Impossible to have sub-zero history or savings in this context

6. TIMELINESS

Potentially outdated snapshots (1.0%)

- **annual_salary** keys indicate potentially outdated intake forms
- Stale financials risk skewing the credit scoring algorithm

Ensuring Quality via Rule-Based Remediation

1. UNIQUENESS

Identity risks isolated

- Duplicate IDs deduplicated (kept most recent submission)
- SSN conflicts quarantined entirely due to unverified identity overlap

2. COMPLETENESS

Strict drops & nulls applied

- Dropped the 6 records lacking required KYC (SSNs/DoB)
- Missing zip codes resolved implicitly (tied to dropped SSN ghost records).
- Missing/invalid emails explicitly nulled (schema compliant)

3. CONSISTENCY

Data formats standardized

- Gender abbreviations mapped to full words ('M' -> 'Male')
- Transform string annual_income to integers

4. VALIDITY

Parsed & verified securely

- Segment-magnitude rule parser resolved 156 dates (ambiguous cases defaulted to DD/MM/YYYY)
- Invalid emails nulled via regex

5. ACCURACY

Strict financial integrity enforced

- Negative financials quarantined rather than imputed

Rationale: Altering financial data without verification introduces severe compliance risks

6. TIMELINESS

Schema locked & flagged

- annual_salary keys renamed to annual_income

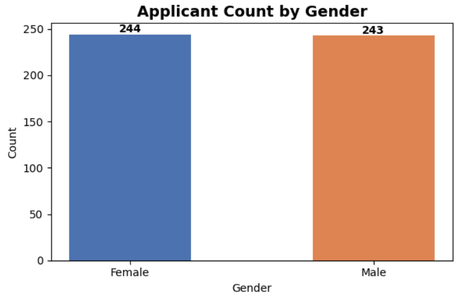
- 487 clean records retained
- 15 records isolated in JSON file
- Locked via MongoDB Schema

Bias Detection & Proxy Discrimination Analysis

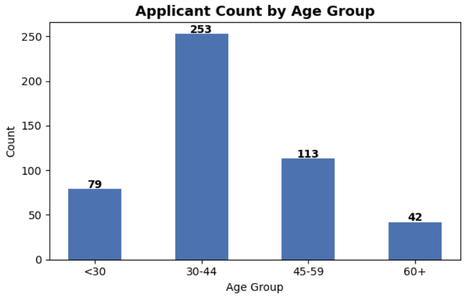
Bias Detection

GENDER AND AGE BIAS

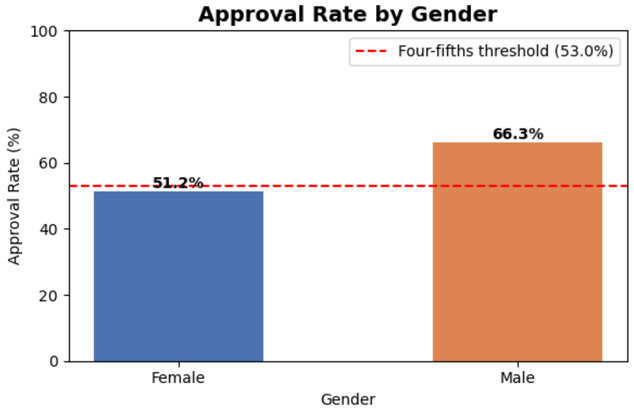
58.7% Approval baseline against which all group disparities are measured



- **Balanced** split
- (Male: 243, Female: 244)

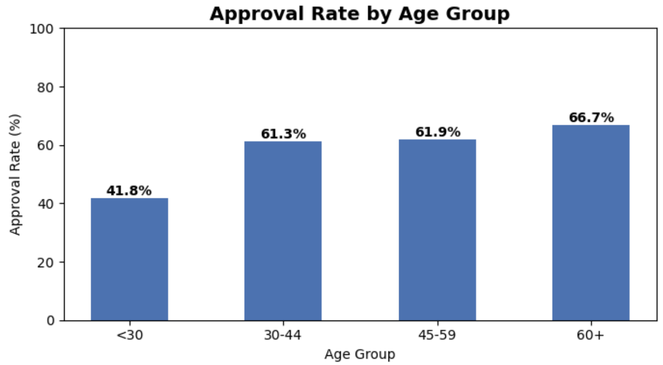


- **30–44 dominates** (253 records)
- 60+ has only 42, making results for that group statistically limited



- **DI of 0.773**, below the 0.8 (four-fifths rule)
- confirmed **statistically significant** (χ^2, p = 0.001)
- **DPD** confirms **15% absolute gap** in favour of male applicants

✓ **Gender Bias**



- **Applicants under 30** are approved at **only 41.8%, compared to 60–66%** for all older groups
- a statistically **significant gap** (χ^2, p = 0.008)

✓ **Age Bias**

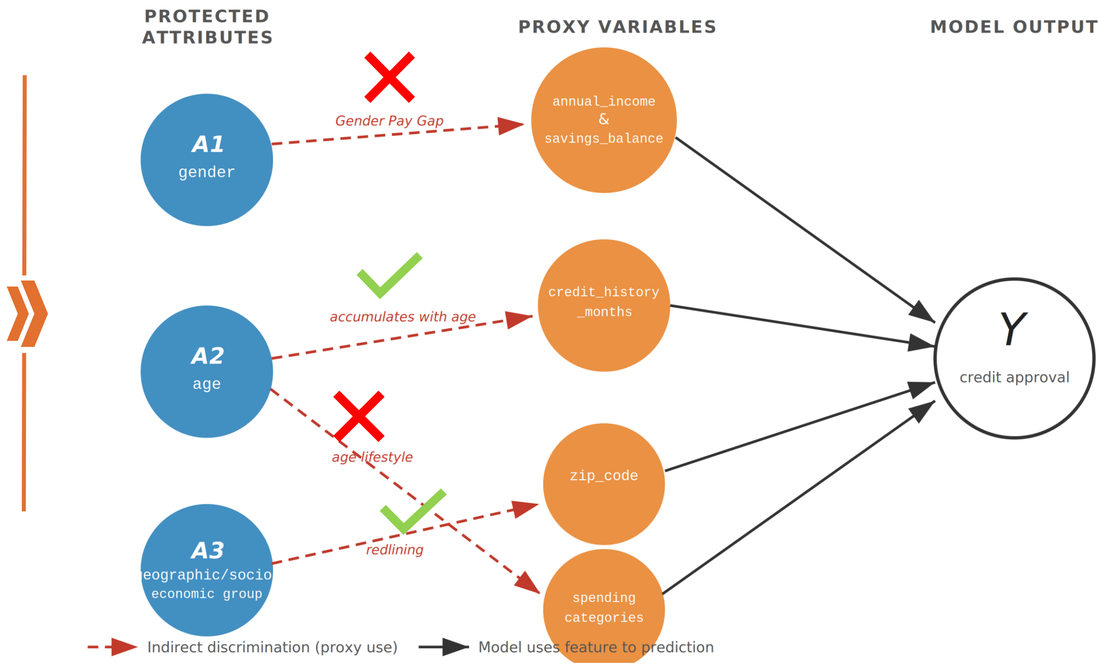
DI: Disparate Impact DPD: Demographic Parity Difference

Proxy Discrimination

GENDER AND AGE DISCRIMINATION

Proxy discrimination occurs when a **seemingly neutral attribute** is strongly correlated with a protected characteristic and the outcome. Hence, the Model is still discriminating, but just indirectly. We investigated four candidate proxies:

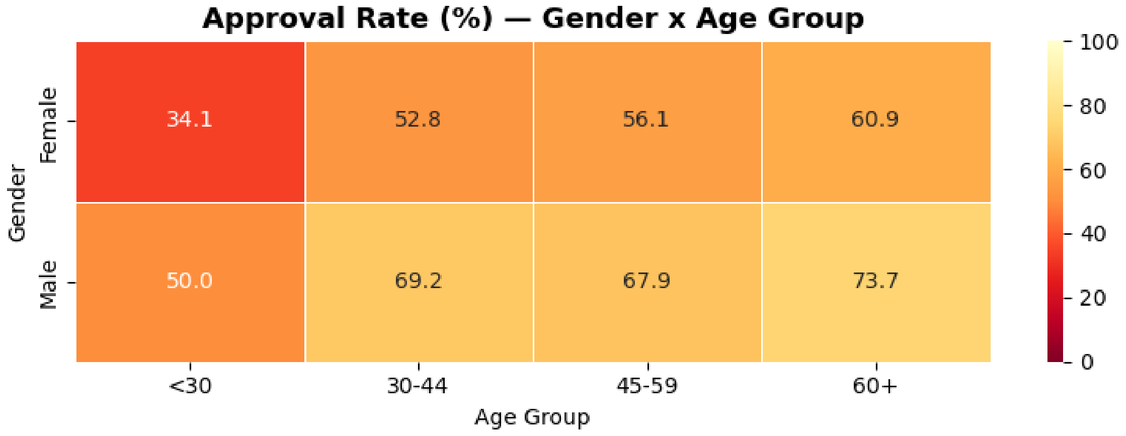
Proxy	Type	Reason	Findings
Annual income/ Savings balance	**Gender**	Financial attributes strongly correlated with gender	**No significant** income/ savings **gap** is detected.
Credit history	**Age**	Older applicants accumulate more credit history by definition	A **strong positive correlation** confirms that older applicants accumulate significantly more credit history.
Zip Code	**Geo-graphic**	Geographic area used as proxy for geographic /socioeconomic groups (redlining)	NYC-area applicants (100xx) are approved at 64.3% versus 52.0% for LA-area applicants (902xx), a statistically **significant difference** (p = 0.025).
Spending Categories (Alcohol, Travel)	**Age**	Spending patterns as age lifestyles	Rent spending has the strongest negative correlation with approval (−0.094), but there are **no spending categories with significant age-group variation**



Interaction Effect

INTERSECTIONAL DISCRIMINATION

Bias can compound when protected attributes **intersect**. We examine the joint effect of **gender** × **age group** on approval rates, to examine **intersectional discrimination**.



We also computed the **DI ratio separately** within each age subgroup to detect whether the gender **gap widens or narrows across age brackets**.

```
Disparate Impact ratio by age subgroup:
  <30: DI = 0.683  (Female 34.1% vs Male 50.0%) <- FAIL
  30-44: DI = 0.763  (Female 52.8% vs Male 69.2%) <- FAIL
  45-59: DI = 0.827  (Female 56.1% vs Male 67.9%)
  60+: DI = 0.826  (Female 60.9% vs Male 73.7%)
```

The **bias compounds** at the intersection of gender and age, with **women under 30 facing the harshest outcomes at only 34.1% approval** (15.9% less than male with DI = 0.683), while the **gender gap narrows for older applicants**

The disadvantage for young women **cannot be explained by gender or age bias alone.**

Privacy & Governance Analysis

Governance Perspective

MAIN TARGETS

Evaluate novacred's credit system from a **Privacy and Compliance Perspective**

Mapp to **gdpr requirements** and **eu AI act**obligations

Identify **privacy risks and governance gaps**in the dataset

GDPR COMPLIANCE RISKS

KEY **GDPR CONCERNS** IDENTIFED

DATA MINIMIZATION
- Detailed behavioural spending data
- Storage of full personal data

LAWFULNESS & TRANSPARENCY
- No consent or lawful basis tracking
- Individuals may not be aware of how their data is being used

STORAGE LIMITATION
- No defined data retention policy

SECURITY OF PROCESSING
- Sensitive identifiers stored in plain text

EU AI ACT

KEY **GOVERNACE** CONCERNS

01.

LACK OF TRANSPARENCY IN AUTOMATED DECISIONS

Rejections only labeled with "algorithm_risk_score"

02.

NO AUDIT TRAIL

No explanation on how decisions are made

03.

NO DOCUMENTED OF HUMAN OVERSIGHT

To registered evidence of human intervention in credit decision

Credit scoring systems = **High-Risk AI** systems

RECOMENDATIONS

FROM A **GOVERNANCE** PRESPECTIVE:

- **IMPROVE DATA PROTECTION**
  - PSEUDONYMIZE PERSONAL IDENTIFIERS (SSNS)
  - ENCRYPT SENSITIVE PERSONAL DATA
- **STRENGTHEN GDPR COMPLIANCE**
  - IMPLEMENT CONSENT/ LAWFUL-BASIS TRACKING
  - DEFINE CLEAR DATA RETENTION POLICY

- **IMPROVE AI GOVERNANCE**
  - IMPLEMENT DECISION LOGGING AND EXPLAINABILITY
  - HUMAN OVERSIGHT FOR CREDIT DECISIONS SHOULD BE IMPLEMENTED

Key Results

FROM THE **DATA ENGINEER**

- **Data Quality Assured:** Refined 502 raw applications down to 487 valid records, remediating all 6 core dimensions.
- **Zero Data Loss:** No records permanently deleted; 15 unresolvable records quarantined for manual review.
- **Pipeline Governance established:** The corrected dataset is securely locked under strict MongoDB **schema validation**.

FROM THE **DATA SCIENTIST**

- **Gender DI** = **0.773** → fails four-fifths rule (15pp gap favouring men, p = 0.001)
- **Age bias**: under-30s approved at **41.8%** vs 60–67% for older groups (p = 0.009)
- **Confirmed proxies:** credit_history_months (r = 0.65 with age) and zip_code (NYC 64.6% vs LA 52.5%)
- **Intersectional** worst case: **women under 30 at 34.1%** approval (DI = 0.683)

FROM THE **GOVERNANCE OFFICER**

- **4 Sensitive PII** fields stored in **plain text** (incl. SSNs) → Critical risk
- **GDPR compliance gaps:** No lawful basis tracking, no retention policy, no audit trail
- **Violation GDPR Art. 22:** Rejections logged only as "algorithm_risk_score"
- System qualifies as **high-risk AI under EU AI Act** → human oversight required